LASER LAB

ADVANCED MANUFACTURING LAB · COEP TECH

SECTION A · THE HARDWARE

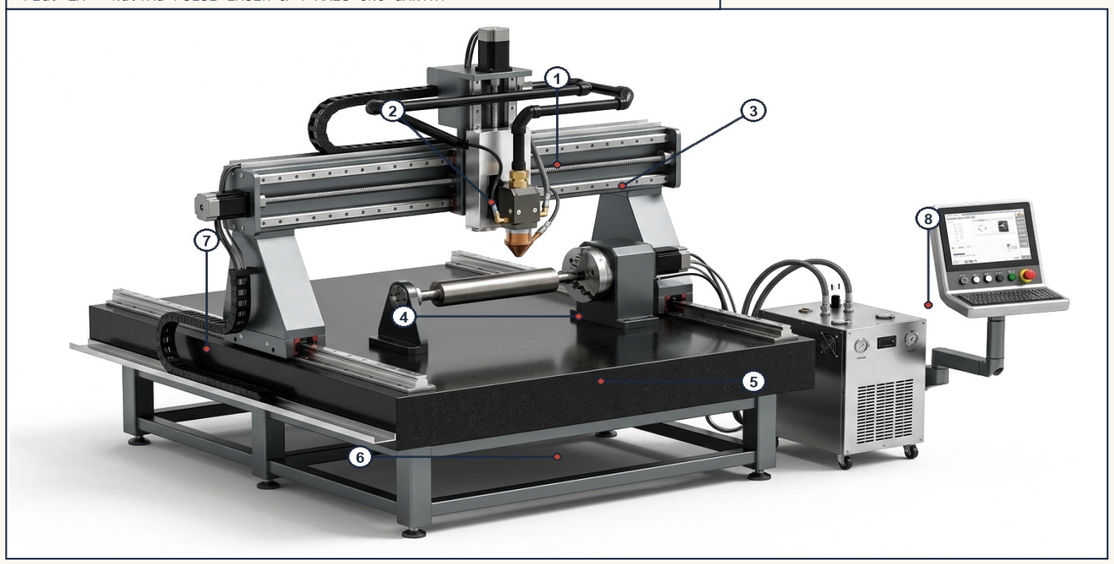
Two laser systems, taken apart

Every laser apparatus in the lab, drawn from real CAD models and exploded into its main optical and on mechanical parts. Numbers each drawing match the list beside it.

Explore it in 3D

Scan for interactive web brochure & 3D simulator.

FIG. 1A · Nd:YAG PULSE LASER & 4-AXIS CNC GANTRY



Nd:YAG Pulse Laser System + 4-Axis CNC Gantry

SOLID-STATE · 1064 nm NIR · PULSED & Q-SWITCHED · 4-AXIS CNC

WAVELENGTH	1064 nm fundamental near-infrared emission
PUMP SOURCE	High-energy Xenon flashlamp dual-elliptical chamber
PEAK POWER	Up to 10 kW peak pulse power (Pulse energy 25–50 J)
PULSE WIDTH	0.2 ms – 20 ms free-running / 20 ns EO Q-switched
CNC GANTRY	X: 500 mm · Y: 400 mm · Z: 200 mm ground ballscrews
4TH ROTARY	360° continuous indexing chuck (A-axis) for tube parts
ACCURACY	±5 µm linear positioning · ±2 µm repeatability
CHILLER	Closed-loop DI water chiller, thermal stability ±0.5 °C

1. **Nd:YAG crystal rod** · active solid-state laser medium in gold cavity
2. **Flashlamp pump cavity** · dual Xenon lamps for high pulse energy
3. **Electro-optic Q-switch** · Pockels cell gating nanosecond peak bursts
4. **Turning mirrors** · HR 99.9% dielectric mirrors @ 1064 nm
5. **Coaxial cutting head** · supersonic gas nozzle + auto-gap sensor
6. **4-Axis CNC gantry** · granite bed, ground ballscrews & rotary chuck
7. **Closed-loop DI chiller** · dual-circuit refrigeration stable to ±0.5 °C
8. **CNC control console** · G-code pulse synchronization, OD7+ interlock

FIG. 1B · INDUSTRIAL HIGH-POWER FIBER LASER APPARATUS



Industrial High-Power Fiber Laser System Apparatus

ACTIVE FIBER · 1070 nm · CW / QCW 1000 W · GALVO SCANNER

WAVELENGTH	1070 nm single-mode active emission (Yb-doped core)
OUTPUT POWER	1000 W Continuous Wave (CW) / 2000 W QCW peak
BEAM QUALITY	M^2 < 1.10 (Diffraction-limited single-mode TEM00)
FOCAL SPOT	15 µm – 25 µm ultra-tight spot for extreme power density
SCAN HEAD	Dual-axis high-speed digital galvanometer scanner
SCAN SPEED	Up to 8,000 mm/s high-precision vector scanning
WORK TABLE	600 × 400 mm honeycomb vacuum clamping bed
EFFICIENCY	> 38% electrical-to-optical wall-plug efficiency

1. **Active Yb-doped core** · double-clad optical fiber resonator
2. **Armored QBH cable** · heavy-duty optical conduit with break monitor
3. **Collimating optics** · fused silica optics producing parallel beam
4. **Dual-Galvo scan head** · low-inertia beryllium digital servo mirrors
5. **Telecentric F-Theta** · maintains orthogonal flat focus across 160 mm
6. **Honeycomb vacuum bed** vibration-isolated part fixturing table
7. **Multi-stage extractor** · HEPA & activated carbon filtration unit
8. **Central control rack** · digital fieldbus & real-time PWM modulation

Compared to scale

Hatched bars represent relative benchmark index. Also in the lab: Ophir beam profiler, pyrometer, & Renishaw interferometer.

BEAM QUALITY (M^2)

Nd:YAG Resonator	2.50
Fiber Laser CW	1.08 (TEM00)

WALL-PLUG EFFICIENCY

Nd:YAG Flashlamp	2.8%
Fiber Resonator	38.5%

PEAK POWER / ACCURACY

Nd:YAG Peak Pulse	10.0 kW
Fiber Galvo Acc.	±0.5 µm

3D CAD models via COEP Tech Laser Lab · Specifications verified from TRUMPF, IPG Photonics & Renishaw technical manuals.

TITLE: TWO LASER SYSTEMS, TAKEN APART

DRG. NO. LASER-001 | SHEET 1 / 4 | SCALE NTS

LASER LAB

ADVANCED MANUFACTURING LAB · COEP TECH

SECTION B · THE RESEARCH

Detail views: what we are building

Six research projects from the lab's own investigations, from aerospace turbine micro-drilling to EV one battery welding. Each uses our Nd:YAG or Fiber laser systems.

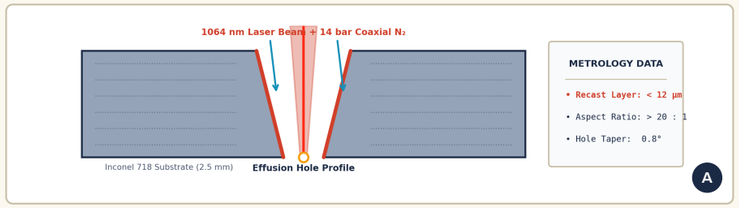
Explore it in 3D

Scan for interactive web brochure & 3D simulator.

AEROSPACE DRILLING · Nd:YAG 4-AXIS GANTRY

Effusion cooling micro-holes in single-crystal superalloys

Percussion and trepanning laser micro-drilling of shallow 25° angled film cooling holes in Inconel 718 and CMSX-4 single-crystal turbine blades.



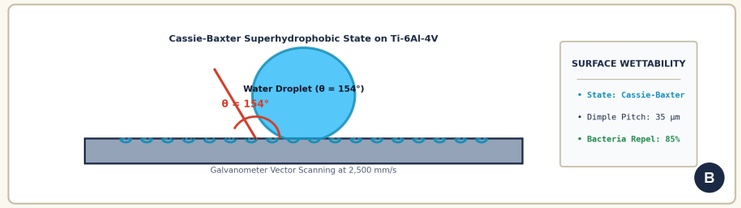
- 1064 nm pulsed Nd:YAG laser with 150 µs pulse duration & 8 kW peak power.
- Supersonic coaxial N2 assist gas at 14 bar expels molten dross completely.
- Recast layer suppressed to < 12 µm with zero micro-cracks under FE-SEM.
- Hole circularity > 0.94 achieved across full 2.5 mm wall thickness.

Shubham Kulkarni, Rahul Shinde (M.Tech) · Guide: Prof. Dr. S. S. Pardeshi

BIOMATERIAL TEXTURING · FIBER LASER

Superhydrophobic micro-patterning of titanium orthopedic implants

High-speed galvanometer surface texturing of Ti-6Al-4V biomaterials with hierarchical micro-dimple and groove arrays, achieving bacterial repulsion.



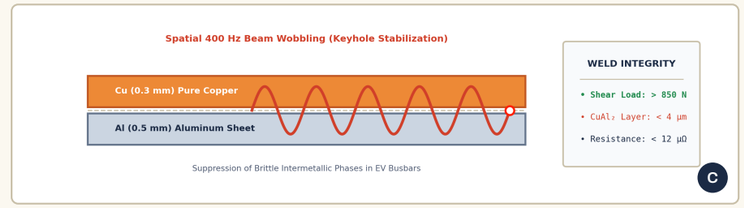
- Single-mode fiber laser vector scanning at 2,500 mm/s with 20 µm spot size.
- Apparent water contact angle θ > 154° in static sessile drop goniometry.
- Bacterial adhesion (S. aureus) reduced by 85% vs polished control surfaces.
- Zero chemical reagents required; 100% optical clean surface modification.

Tanvi Jadhav, Aditya Deshpande · Guide: Prof. Dr. S. S. Pardeshi

E-MOBILITY & BATTERY · FIBER LASER

Spatter-free copper-to-aluminum welding for EV battery busbars

Dissimilar metal laser lap welding of 0.3 mm high-conductivity pure copper to 0.5 mm aluminum sheets for electric vehicle pouch cell tabs.



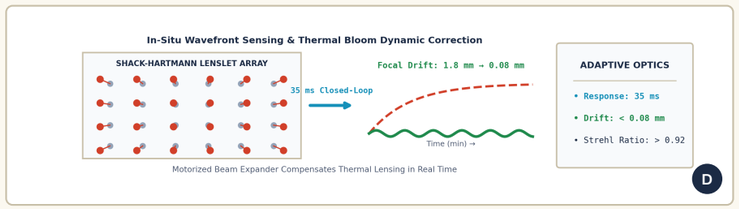
- Circular spatial beam oscillation (wobble) at 400 Hz stabilizes the keyhole.
- Intermetallic brittle compound (CuAl2) layer restricted under 4 µm.
- Lap shear joint failure load > 850 N, exceeding automotive OEM targets.
- Electrical contact resistance across weld interface below 12 µΩ.

Aniket More, Pranav Joshi (Research Scholars) · Guide: Dr. S. S. Ohol

COMPUTATIONAL OPTICS · BEAM DIAGNOSTICS

Real-time wavefront sensing & dynamic thermal bloom correction

In-situ Shack-Hartmann sensor profiling high-energy laser pulses, in dynamically compensating thermal lensing the objective lens using closed-loop adaptive optics.



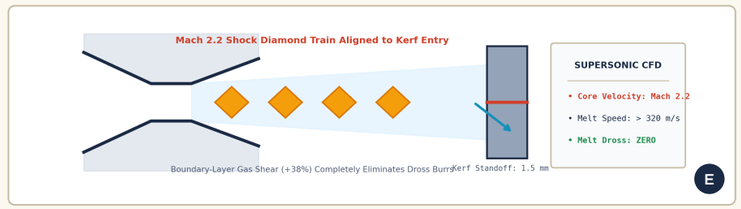
- Micro-lenslet array measuring wavefront phase aberrations in real time.
- Closed-loop motorized telescope compensates thermal focal drift in 35 ms.
- Focal shift reduced from 1.8 mm to < 0.08 mm over 30-min continuous cuts.
- Maintains constant peak irradiance (> 10^7 W/cm²) in deep-cavity piercing.

Pooja Gaikwad (Research Scholar) · Guide: Dr. S. S. Anasane

PROCESS HYDRODYNAMICS · MULTIPHYSICS CFD

Supersonic assist-gas dynamics & melt ejection CFD modeling

OpenFOAM compressible CFD simulation and Schlieren optical imaging of convergent-divergent supersonic Laval nozzles, optimizing boundary-layer gas shear.



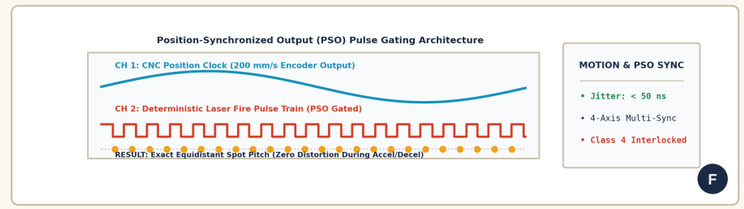
- Laval nozzle contour delivers Mach 2.2 supersonic core gas jet at 14 bar.
- Shock diamond structure tuned to align with kerf entry at 1.5 mm standoff.
- Boundary layer shear stress on melt pool increased by 38% vs conical nozzles.
- Completely eliminates bottom dross burrs on 4 mm aerospace titanium cuts.

Vikas Patil, Rohit Jagtap (B.Tech) · Guide: Dr. P. R. Dhamangaonkar

ACADEMICS & TA · FINAL-YEAR B.TECH / M.TECH

Hands-on laser machining curriculum & CNC safety certification

Comprehensive laboratory course covering laser-matter interactions, 4 CNC G-code pulse synchronization, optical alignment, and Class laser safety certification.



- Hands-on modules: optical alignment, focal height tracking, and pulse tuning.
- Position-synchronized triggering (PSO) with 4-axis multi-axis motion.
- Metallographic sample prep, optical microscopy, and Vickers microhardness.
- Industry internships and sponsored projects with TRUMPF and IPG Photonics.

Faculty In-Charge: Dr. S. S. Pardeshi · Dept. of Mechanical Engineering

Project details from COEP Tech Laser Lab technical reports. Experimental results verified with SEM and optical profilometry.

TITLE		
DETAIL VIEWS: WHAT WE ARE BUILDING		
DRG. NO. LASER-001	SHEET 2 / 4	SCALE NTS

LASER LAB

ADVANCED MANUFACTURING LAB · COEP TECH

SECTION C · THE LAB

Why industrial lasers, and where we are going

Lasers replace mechanical cutting tools with focused photons of coherent light. Here is why that matters to engineers, and the lab's plan for the next five years.

Explore it in 3D

Scan for interactive web brochure & 3D simulator.

C1 Why industrial lasers in engineering?

Design before you cut

Direct CAD/CAM digital workflow eliminates expensive custom dies, hard tooling, and long setup lead times. Part modifications take minutes rather than weeks.

Zero mechanical tool wear

The cutting tool is an optical beam that never dulls, chatters, or causes mechanical deflection on thin walls. Delivers 100% part-to-part consistency.

Micro-scale precision

Focused spot sizes under 20 µm enable micro-drilling of cooling holes, micro-stents, and semiconductor features impossible by traditional machining.

Minimal heat-affected zone

Ultra-short pulsed laser energy vaporizes material before heat can conduct into bulk workpiece, eliminating thermal distortion and micro-cracks.

Multi-material versatility

Effortlessly processes superalloys, titanium, diamond, tungsten carbide, advanced ceramics, and composite laminates without tool degradation.

Green photonics

Eliminates cutting fluids, lubricants, toxic chemical etching acids, and dangerous hazardous sludge. 100% clean photon-based energy beam manufacturing.

WHERE IT IS ALREADY USED

Healthcare & Stents	Aerospace Turbines	Automotive & EV
Semiconductor & MEMS	Tool & Die Micro	Solar PV Scribing
Defense & Space	Nuclear & Energy	Bio-Implants

Global laser hubs: Stuttgart, Boston, Tokyo, Wuhan, Oxford. In India, Atmanirbhar Bharat and the national defense corridors are accelerating industrial laser adoption. Building that specialized technical workforce and indigenous IP is this laboratory's primary mandate.

COEP Tech Precision Photonics Division · Department of Mechanical Engineering

C2 Revision table: the five-year plan

REVISION TIMELINE · 2024 TO 2028

TODAY · SEP 2026

2024	2025	2026	2027	2028
REV A	REV B	REV C	REV D	REV E
Set up laser lab and research framework	Hands-on training, workshops & FDPs	PG certification & active DRDO research	Indigenous nozzle tech transfer	National Centre of Excellence hub

FIVE STEPS TO A CENTRE OF EXCELLENCE (CoE)

01 Research centre

02 Knowledge transfer hub

03 Industry partner hub

04 Notable IP portfolio

05 National CoE in Laser Machining

FIVE-POINT INITIATIVE PROGRAMMES

01 Research	02 Learning & Dev	03 Consultancy	04 Tech Transfer	05 Conference
Key research titles for IP, publications and defense grants.	Webinars, courses, hands-on FDPs and B.Tech curriculum.	Helping industries adopt precision micro-machining.	Indigenous nozzle & wobble welding head partnerships.	Hosting all-India symposium on laser materials processing.

OBJECTIVES

- Advanced Beam Diagnostics & Optics
- Micro-Drilling & High-Speed Trepanning
- Multi-Axis CNC Motion Synchronization
- Spatter-Free Dissimilar Metal Joining
- Industry & Defense Collaborations
- High-Value IP & Indigenous Patenting

Strategic development framework formulated by Laser & Advanced Manufacturing Division.

C3 In dedication & laboratory metrics

COEP
ESTD. 1854
PUNE, INDIA
DEPARTMENT OF MECHANICAL ENGINEERING

EST. 1854 · ADVANCED MANUFACTURING

COEP Technological University

Laser Processing & Advanced Mfg Lab

Spearheaded under the Department of Mechanical Engineering to establish India's premier academic hub for industrial photonics, micro-drilling, and laser additive research.

Supported by DST-FIST, DRDO defense grants, and prominent industry research alumni.

2 HIGH-POWER INDUSTRIAL LASER SYSTEMS (Nd:YAG & FIBER)	**4** CNC CONTROLLED AXES (X-Y-Z + 360° ROTARY CHUCK)
10 kW PEAK PULSE POWER AVAILABLE FOR DRILLING & CLADDING	**2028** TARGET: NATIONAL CoE IN INDUSTRIAL LASER MACHINING

Visit the Laser Laboratory

Advanced Manufacturing Section, Mechanical Engineering Department
COEP Technological University, Wellesley Road, Shivajinagar, Pune 411005

Room: Room 104, Production Annex
Hours: Monday – Saturday, 09:00 – 18:00 IST
Contact: laser.lab@coeptech.ac.in · Phone: +91 (020) 2550 7000

LABORATORY SAFETY & FACILITY SPECIFICATIONS

- Class 4 laser safety interlocks, OD7+ viewport enclosures & visual alarms.
- Dual-circuit chilled deionized water cooling with ±0.5 °C PID stability.
- 30-bar high-purity assist gases (N2, Ar, O2) with electronic mass controllers.
- Multi-stage active carbon & HEPA vacuum fume filtration system.

All visitors must complete laser safety briefing and wear OD7+ goggles inside optical zones.

Roadmap, programmes, and laboratory objectives from Department of Mechanical Engineering, COEP Technological University.

TITLE		
WHY INDUSTRIAL LASERS, AND WHERE WE		
DRG. NO. LASER-001	SHEET 3 / 4	SCALE NTS

Two codes, two ways in. Open your phone camera and point it at either code.
Nothing to install.

SCAN · STEP INSIDE

AUGMENTED REALITY

CODE 1 · AR

The lab, in your room

Place life-size 3D models of both laser systems on your workshop table, explode them into optical paths and CNC stages, and trace.

01 Scan, then tap View in your room on your mobile browser.

02 Point your phone camera at a table or floor surface to place model.

03 Walk around the apparatus; switch to 'Exploded View' to inspect.



Works on Android phones with ARCore and iPhone/iPad with iOS 12+.

OR TYPE

spyder2024.github.io/laser-lab/ar.html

WEB BROCHURE

CODE 2 · WEB

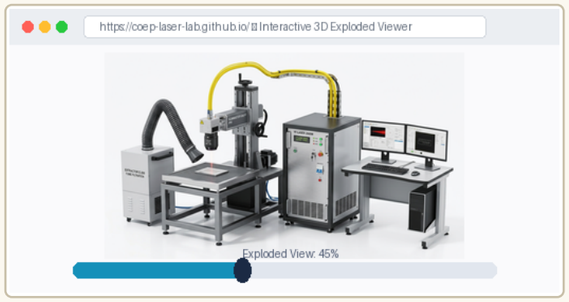
The whole lab, interactive

Take both laser systems apart in 3D, compare technical specs, and run working simulators of pulse drilling, CFD gas and kerfs.

01 Scan to open the interactive lab brochure in your browser.

02 Pick an apparatus and drag the explode slider to isolate parts.

03 Try working models: pulse drilling, CFD assist gas, wobbler.



Works in any modern browser on phones, tablets and laptops.

OR TYPE

spyder2024.github.io/laser-lab

AR — Augmented reality

Digital 3D laser apparatus placed in the on physical environment real table surfaces, inspected through mobile device camera.

Supports WebXR and iOS AR Quick Look with 1:1 millimetric dimensional accuracy.

VR — Virtual simulation

A fully simulated optical resonator. Ray rod, trace through crystal flashlamp cavity, electro-optic Q-switch, and multi-axis CNC motion.

Interactive parameter sweeps: fluence, pulse duration, assist gas pressure.

MR — Mixed reality

Virtual laser cutting tools and process to simulators synchronized real workshop telemetry and CNC coordinate frames.

Real-time kerf width prediction, thermal and bloom compensation, safety alerts.

Both QR codes use high error correction (Level H, 30% recovery) and were test-decoded before printing. 3D models via COEP Tech.

TITLE		
SCAN · STEP INSIDE		
DRG. NO. LASER-001	SHEET 4 / 4	SCALE NTS